Homework 8

Due Friday 03/13 at 5 p.m.

Problem 1. Pole Placement Design using CCF (25 Points)

Consider the system

$$G(s) = \frac{1}{(s^2 + 2s + 100)(s^2 + 0.25s - 5)}$$

(a) Convert the transfer function to a CCF state space representation. Report the A, B, C and D matrices.

(b) Plot the poles and zeros of the system on a complex plane. Qualitatively characterize how each pole will contribute to the response of the system. (You may use MATLAB).

(c) Select a desired set of poles that will yield

(i) a percent overshoot of less than 5%

(ii) a settling time of no more than 4.6 seconds

Hint: You will only get two poles to match your transient specs. You need to choose a spot for all closed-loop poles

(d) Use pole placement to design a set of gains that will place your closed-loop poles at the locations you have selected. Show all work (i.e. do NOT use MATLAB functions to compute the gains).

(e) Use MATLAB to validate that the eigenvalues of your closed-loop system match the desired pole locations. Show the MATLAB outputs.

(f) Show in MATLAB that your design meets the transient specs by simulating a step response.

Problem 2. Infinite Horizon LQR Design and Tuning (25 Points)

Consider the following system that represents the dynamics of a simple inverted pendulum linearized about it's vertical position.

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ \omega_0^2 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ -1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & 1 \end{bmatrix} x$$

The state vector x consists of angular displacement (x_1) and angular velocity ($x_2 = \dot{x}_1$). Assume $\omega_0 = 1$ rad/s and the initial state $x(0) = \begin{bmatrix} 1/2 & 1 \end{bmatrix}^T$.

(a) Using LQR design methods, design a gain matrix that equally weights all states and inputs. Report your gain K. Plot the state response of your system subject to the control law $u = -Kx$, not subject to any reference. Separately, plot the applied control input under the designed controller. *Hint: Use MATLAB's* `icare` *function to solve the infinite horizon Ricatti equation.*

(b) The course assistants are impatient, and want the state response to settle 3x faster. Tune your Q and R matrices in order to achieve this. Justify your choices, report your design iterations, and include plots/simulations when relevant. Finally, report the Q, R, and K matrices corresponding to your final design.

(c) The course assistants realize that the pendulum controller does not have infinite power. So, they now impose an upper bound on your control at $u_{upper} = 4$. Tune your Q and/or R matrices such that this requirement is satisfied while attempting to maintain low response times (≈2x your response time from b)). Plot your state response and control input.

(d) Assume that the frequency, ω_0 of the system varies by ±20%. Simulate your closed-loop state response using your gain matrix K from parts (b) and (c) at different values of ω_0. Submit two plots, one for each controller you designed, each having multiple state responses for varying ω_0. Comment on your findings.

Problem 3. Infinite Horizon LQR with Output Penalization (25 Points)

Consider the LTI system $\dot{x} = Ax + Bu$ and $y = Cx$. Traditionally the cost function that is optimized for infinite time horizon LQR design is

$$J = \int_0^\infty \big(x(t)^T Q x(t) + u(t)^T R u(t)\big) dt$$

You have seen in class how the gain K that is used in linear feedback (i.e. $u = -Kx$) is derived from minimizing this subject to the linear dynamics of the system. It is sometimes desirable to alter the cost function to achieve different goals.

(a) Provide the Riccati equations and an expression for gain K for the following objective function which seeks to penalize the output y

$$J = \int_0^\infty \big(y(t)^T Q y(t) + u^T(t) R u(t)\big) dt$$

(b) Now consider the system

$$A = \begin{bmatrix} -4 & 1 \\ -2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Design a gain matrix using LQR methods with output penalization, and report your Q and R matrices along with the resulting K. Ensure your controller does the following

(i) More heavily penalizes the first output vs. the second output

(ii) More heavily penalizes the second input vs. the first input

(iii) Generally penalizes control more than any of the outputs

(c) Plot the time evolution of both states and both outputs. Set your initial condition to $x_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$.

Problem 4. Observer Design (25 Points)

Figure 1 shows a velocity observation system where x_1 is the velocity to be observed. An observer is to be constructed to track x_1, using u and x_2 as inputs. The variable x_2 is obtained from x_1 through a sensor having the known transfer function

$$\frac{2-s}{2+s}$$

as shown in Figure 1.

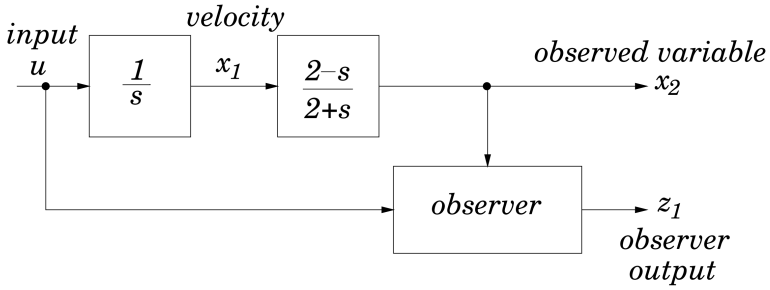


Figure 1: Velocity Observation System.

(a) Derive a set of state-space equations for the system with state variables x_1 and x_2, input u, and output x_2.

(b) Design an observer with states z_1 and z_2 to track x_1 and x_2 respectively. Choose both observer eigenvalues to be at -4. Write out the state-space equations for the observer.

(c) Derive the combined state equation for the system plus observer. Take as state variables x_1, x_2, $e_1 = x_1 - z_1$, and $e_2 = x_2 - z_2$. Take u as input and z_1 as the output. Is this system controllable and/or observable? Give physical reasons for any states being uncontrollable or unobservable.

(d) What is the transfer function relating u to z_1? Explain your result.

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Hint: You will only get two poles to match your transient specs. You need to choose a spot for all closed-loop poles

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(e) Use MATLAB to validate that the eigenvalues of your closed-loop system match the desired pole locations. Show the MATLAB outputs.

(f) Show in MATLAB that your design meets the transient specs by simulating a step response.

a) $G(s) = \frac{1}{(s^2+2s+100)(s^2+0.25s-5)}$

$= \frac{1}{s^2(s^2+2s+100)+0.25s(s^2+2s+100)-5(s^2+2s+100)}$

$= \frac{1}{s^4+2s^3+100s^2+0.25s^3+0.5s^2+25s-5s^2-10s-500}$

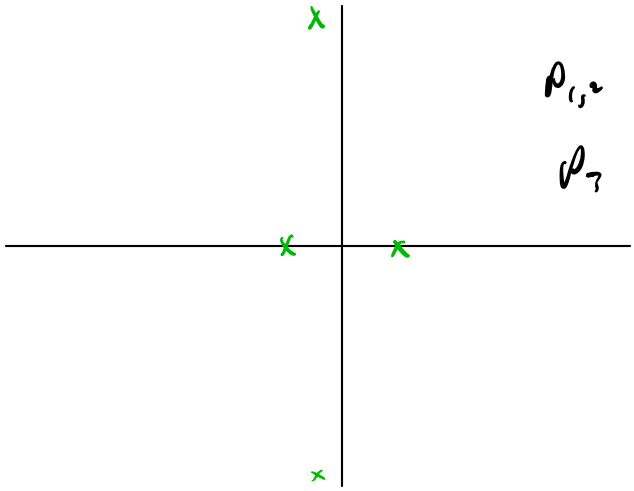
$= \frac{1}{s^4+2.25s^3+95.5s^2+15s-500}$ $\quad p_0=1$

$a_0=-500 \quad a_1=15 \quad a_2=95.5 \quad a_3=2.25$

$$A=\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 500 & -15 & -95.5 & -2.25 \end{bmatrix} \quad B=\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad C=[1000] \quad D=0$$

b) $p_1, p_2 = \frac{-2 \pm \sqrt{4-400}}{2} = -1 \pm j\sqrt{99} = -1 \pm 3.511\,j$

$p_3, p_4 = \frac{-0.25 \pm \sqrt{.0625-4(-5)}}{2} = -0.125 \pm \sqrt{20.064} = -2.3646,\ 2.1146$



$p_{1,2}$ will contribute a decaying sinusoid

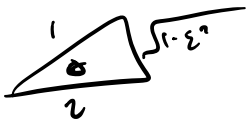
p_3 will contribute a decaying exponential

p_4 will contribute an exponential growth.

NO ZERO

c) putting poles 3,4 at stable locations $\quad M_p \leq 0.05 \quad t_s \leq 4.6s$

$$M_p = e^{\frac{-\pi}{\tan\theta}} \qquad \theta = \tan^{-1}\left(\frac{-\pi}{\ln M_p}\right) = \tan^{-1}\left(\frac{-\pi}{\ln(0.05)}\right) = 46.36$$



$\cos\theta = \zeta \qquad \zeta = 0.6901 \qquad t_s = \frac{4.6}{\sigma} = 4.6 \qquad \sigma = 1$

$\omega_n \zeta = 1 \qquad \omega_n = 1.4491 \qquad \omega_d = \omega_n\sqrt{1-\zeta^2} = 1.0487 \xrightarrow{\text{set at}} 1$

$\text{Poles}_{1,2} = -1 \pm j$ $\qquad$ $\text{Poles}_{3,4} = -100_j - 120$

For $(s+1+j)(s-1-j)$ $\quad i = \sqrt{-1} = 2$

d) $P_d = (s-p_1)(s-p_2)(s+100)(s+120) = (s^2 - (p_1+p_2)s + p_1p_2)(s+100)(s+120)$

$= (s^2+2s+2)(s+100)(s+120) = (s^3+2s^2+2s+100s^2+200s+200)(s+120)$

$= (s^3+102s^2+202s+200)(s+120) = s^4+102s^3+202s^2+200s+120s^3+12240s^2+24240s+24000$

$= s^4 + 222s^3 + 12442s^2 + 12440s + 24000 \qquad \beta_{0-3}$

$P_{ol} = s^4 + 2.25s^3 + 95.5s^2 + 15s - 500 \qquad \alpha_{0-3}$

$k_0 = 24000+500 = 24500 \qquad k_1 = 12440 - 15 = 12425$

$k_2 = 12442 - 95.5 = 12346.5 \qquad k_3 = 222 - 2.25 = 219.75$

$$K = [24500 \quad 12425 \quad 12346.5 \quad 219.75]$$

e)

```
A_bar = A-B*K;
G_new = tf(ss(A_bar,B,C,D));
stepinfo(G_new)
```

struct with fields:

```
      RiseTime: 1.519544629839307
  TransientTime: 4.234576809492349
   SettlingTime: 4.234576809492349
    SettlingMin: 3.796754242138947e-05
    SettlingMax: 4.346639457427998e-05
      Overshoot: 4.319346978271832
     Undershoot: 0
           Peak: 4.346639457427998e-05
       PeakTime: 3.177567428331079
```

```
Pole 1: -120.0000 +0.0000j
Pole 2: -100.0000 +0.0000j
Pole 3:   -1.0000 +1.0000j
Pole 4:   -1.0000 -1.0000j
```

(ii) a settling time of no more than 4.6 seconds

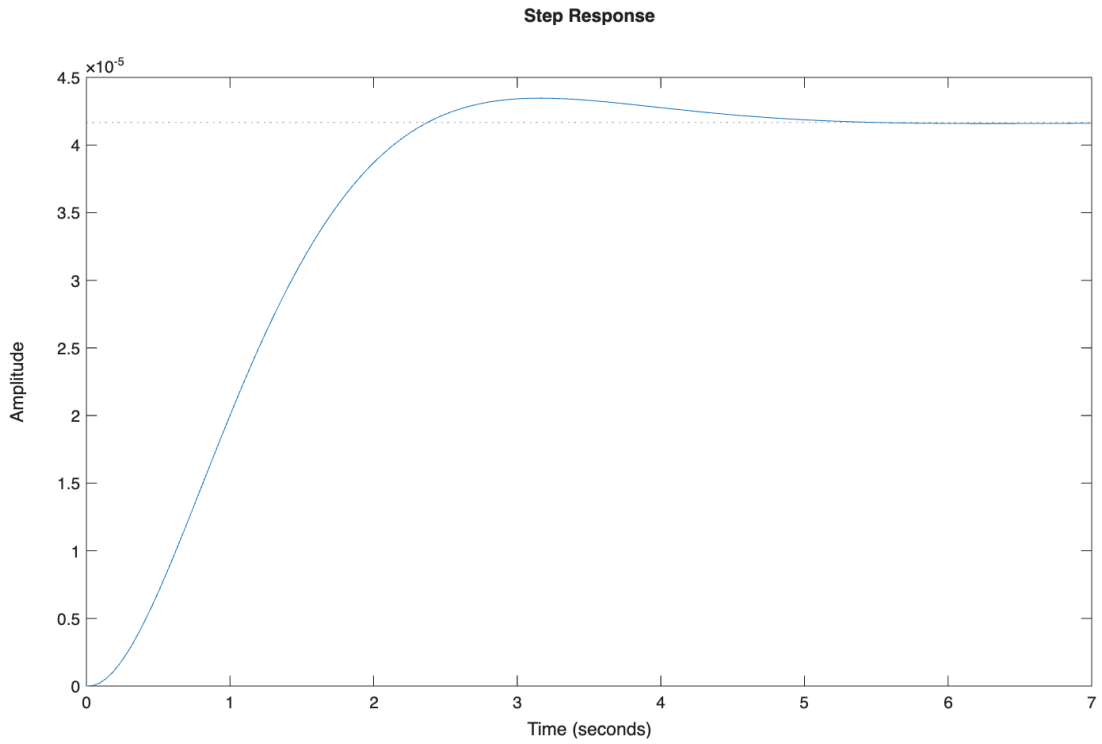
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f)



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a) $A = \begin{bmatrix} 0 & 1 \\ \omega_0^2 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ $C = \begin{bmatrix} 2 & 1 \end{bmatrix}$

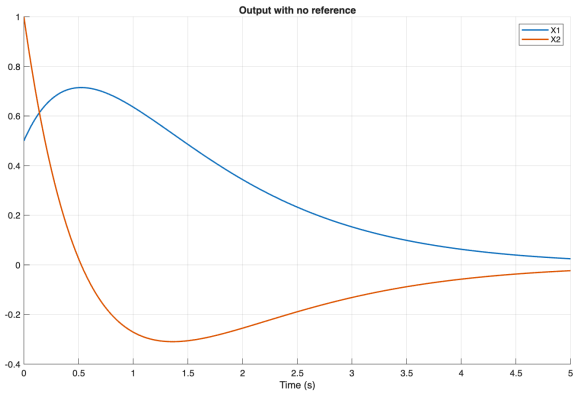
$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $R = [1]$

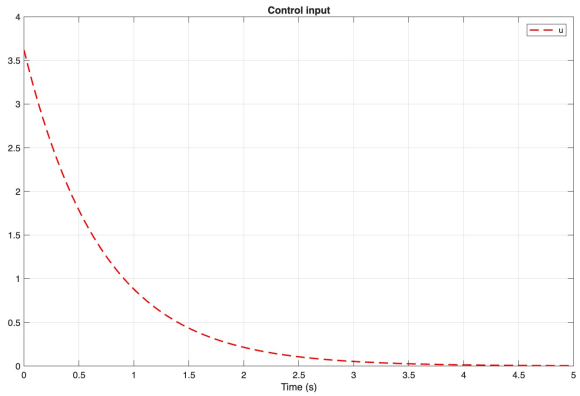
$PA + A^TP - PBR^{-1}B^TP + Q = 0$

Matlab 9.45

$P = \begin{bmatrix} 3.414 & 2.414 \\ 2.414 & 2.414 \end{bmatrix}$

$K = \begin{bmatrix} -2.414 & -2.414 \end{bmatrix}$





struct with fields:

```
      RiseTime: 2.545509074171203
  TransientTime: 4.376883232783682
   SettlingTime: 4.376883232783682
    SettlingMin: -1.414150395437601
    SettlingMax: -1.276148212833486
      Overshoot: 0
     Undershoot: 0
           Peak: 1.414150395437601
       PeakTime: 10.545839725910376
```

b) $u_{max}^2 = \left(\frac{1}{3}\right)^3 = \frac{1}{9} \rightarrow Q = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}$ $R = [1]$ → $K = \begin{bmatrix} -3 & -2.646 \end{bmatrix}$

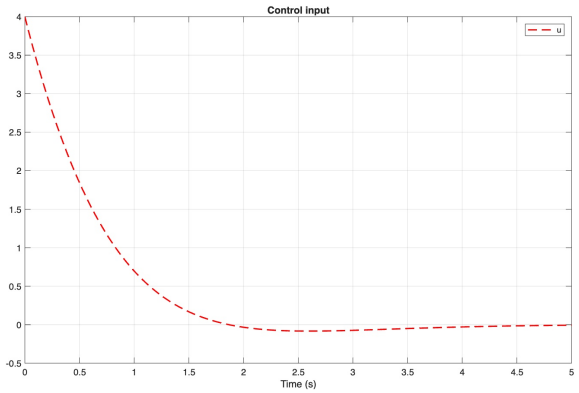
```
      RiseTime: 1.004112918918774
  TransientTime: 1.454907454608652
   SettlingTime: 1.454907454608652
    SettlingMin: -0.642946039612665
    SettlingMax: -0.572328078173599
      Overshoot: 1.658694888043200
     Undershoot: 0
           Peak: 0.642946039612665
       PeakTime: 2.262157706344750
```

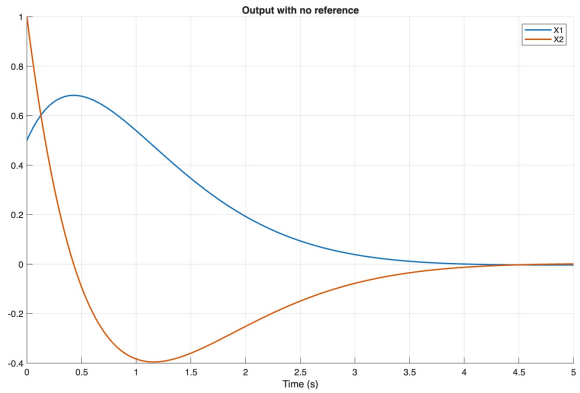
want ≈1.455

c) $u_{max} = 4$ → $Q = \begin{bmatrix} 160 & 0 \\ 0 & 1 \end{bmatrix}$ $R = 16$ $t_s < \sim 2.195$

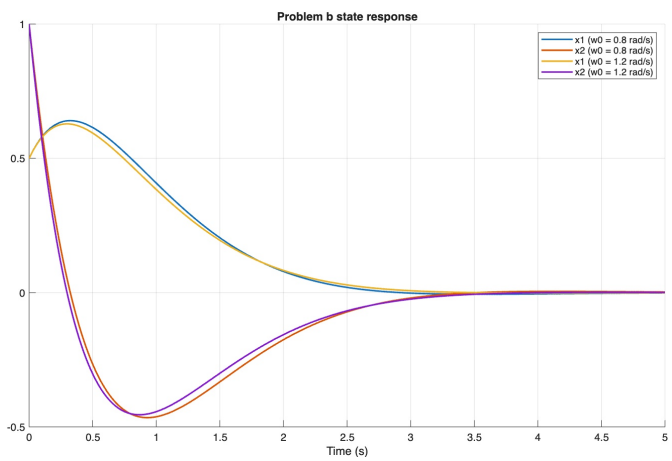
$K = [-3.049 \quad -2.474]$

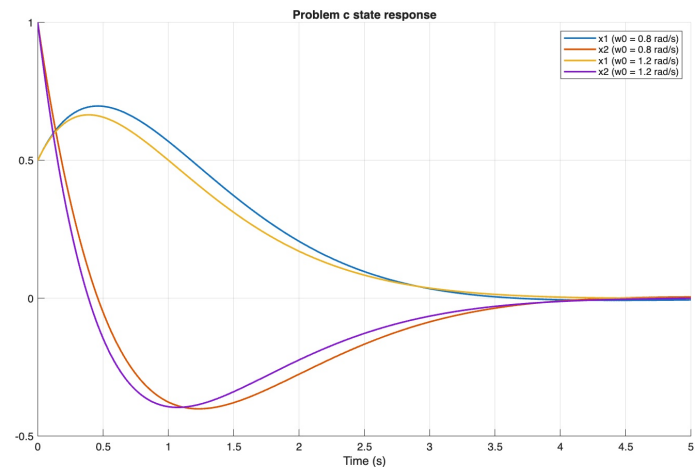
```
RiseTime: 1.465819212965628
TransientTime: 2.177564820340683
SettlingTime: 2.177564820340683
SettlingMin: -0.984463782104994
SettlingMax: -0.885143589069430
Overshoot: 0.877519061002485
Undershoot: 0
Peak: 0.984463782104994
PeakTime: 3.313853129914809
```





d) $Q = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}$ $R = 1$ (Problem b) $Q = \begin{bmatrix} 160 & 0 \\ 0 & 1 \end{bmatrix}$ $R = 50$ (Problem c)





(b) has faster response

Settle time

	part (b)	part (c)
$\omega_0 = 0.8$	2.612s	> 2.025 s
$\omega_0 = 1.2$	1.535s	< 2.257s

part b

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You have seen in class how the gain K that is used in linear feedback (i.e. $u = -Kx$) is derived from minimizing this subject to the linear dynamics of the system. It is sometimes desirable to alter the cost function to achieve different goals.

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$$J = \int_0^\infty (y(t)^T Q y(t) + u^T(t) R u(t))dt$$

(b) Now consider the system

$$A = \begin{bmatrix} -4 & 1 \\ -2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Design a gain matrix using LQR methods with output penalization, and report your Q and R matrices along with the resulting K. Ensure your controller does the following

(i) More heavily penalizes the first output vs. the second output

(ii) More heavily penalizes the second input vs. the first input

(iii) Generally penalizes control more than any of the outputs

(c) Plot the time evolution of both states and both outputs. Set your initial condition to $x_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$.

a)

$$\min J = \min\{ y^T Q_y y + u^T R u \}$$

$$= \min\{ x^T A^T Q_y A x + u^T R u \}$$

$$Q_x = C^T Q_y C$$

save $K = R^{-1} B^T P$

$$A^T P + PA - PBR^{-1}B^T P + C^T Q_y C = 0$$

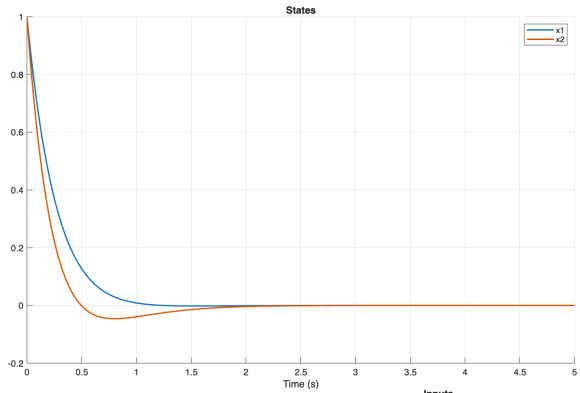
$$K = R^{-1} B^T P$$

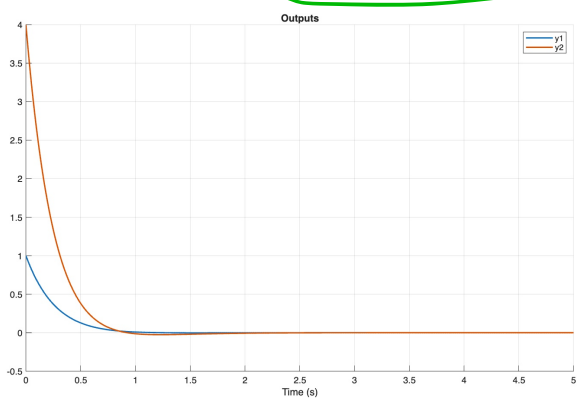
c) $Q_1 > Q_2$ $R_2 > R_1$ $\rho \approx 2$ $x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

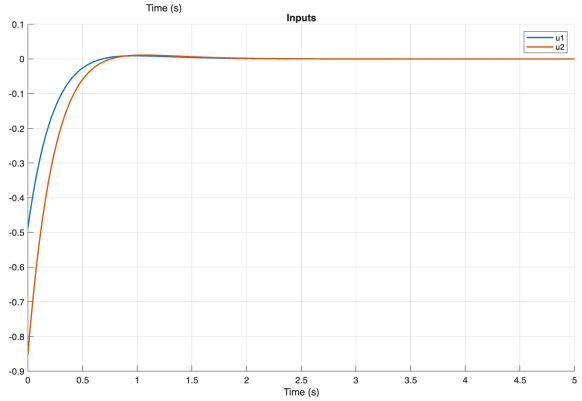
$$Q_y = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad R_1 = 2\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

$$Q_x = C^T Q_y C = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 3 \\ 3 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad R = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \quad K = \begin{bmatrix} 0.210 & 0.276 \\ 0.471 & 0.382 \end{bmatrix}$$







Problem 4. Observer Design (25 Points)

Figure 1 shows a velocity observation system where x_1 is the velocity to be observed. An observer is to be constructed to track x_1, using u and x_2 as inputs. The variable x_2 is obtained from x_1 through a sensor having the known transfer function

$$\frac{2-s}{2+s}$$

as shown in Figure 1.

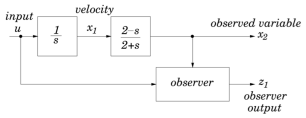


Figure 1: Velocity Observation System.

(a) Derive a set of state-space equations for the system with state variables x_1 and x_2, input u, and output x_2.

(b) Design an observer with states z_1 and z_2 to track x_1 and x_2 respectively. Choose both observer eigenvalues to be at -4. Write out the state-space equations for the observer.

(c) Derive the combined state equation for the system plus observer. Take as state variables x_1, x_2, $e_1 = x_1 - z_1$, and $e_2 = x_2 - z_2$. Take u as input and z_1 as the output. Is this system controllable and/or observable? Give physical reasons for any states being uncontrollable or unobservable.

(d) What is the transfer function relating u to z_1? Explain your result.

$X_2 = \frac{2-s}{2+s} X_1$ $\qquad X_1 = \frac{1}{s} U$

$\dot{X}_1 = U$

$2X_2 + sX_2 = 2X_1 - sX_1$

$2X_2 + \dot{X}_2 = 2X_1 - \dot{X}_1 = 2X_1 - U$

$\dot{X}_2 = 2X_1 - 2X_2 - U$

$\dot{X}_1 = U$

$\dot{X} = \begin{bmatrix} 0 & 0 \\ 2 & -2 \end{bmatrix} X + \begin{bmatrix} 1 \\ -1 \end{bmatrix} U \qquad y = [0\ 1] X$

$A = \begin{bmatrix} 0 & 0 \\ 2 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad C = [0\ 1] \quad D = 0$

5) $\lambda = -4, -4 \qquad A - TC = \begin{bmatrix} 0 & 0 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} [0\ 1]$

$\begin{bmatrix} 0 & 0 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} 0 & T_1 \\ 0 & T_2 \end{bmatrix} = \begin{bmatrix} 0 & -T_1 \\ 2 & -2-T_2 \end{bmatrix}$

$\lambda I - (A - TC) = \begin{bmatrix} \lambda & T_1 \\ -2 & \lambda + T_2 + 2 \end{bmatrix} \qquad \lambda(\lambda + T_2 + 2) + 2T_1 = 0$

$\lambda^2 + (T_2 + 2)\lambda + 2T_1 = 0$

$\lambda_{des} \rightarrow (\lambda+4)(\lambda+4) = \lambda^2 + 8\lambda + 16 = 0$

$T_2 + 2 = 8 \qquad 2T_1 = 16$

$T = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$

$\dot{\hat{x}} = (A - TC)\hat{x} + Bu + Ty \qquad \hat{y} = C\hat{x}$

$$A-Tc=\begin{bmatrix}0 & -T_1\\ 2 & -2-T_2\end{bmatrix} \qquad \hat{A}=\begin{bmatrix}0 & -8\\ 2 & -8\end{bmatrix}$$

$$\dot{\hat{x}}=\begin{bmatrix}0 & -8\\ 2 & -8\end{bmatrix}\hat{x}+\begin{bmatrix}1\\ -1\end{bmatrix}u+\begin{bmatrix}8\\ 6\end{bmatrix}y \qquad \hat{y}=[0\ 1]\hat{x}$$

$$\dot{x}=Ax+Cu \qquad e=x-\hat{x} \rightarrow \qquad \dot{e}=(A-Tc)\,e$$

$$\begin{Bmatrix}\dot{x}\\ \dot{e}\end{Bmatrix}=\begin{bmatrix}A & 0\\ 0 & A-Tc\end{bmatrix}\begin{Bmatrix}x\\ e\end{Bmatrix}+\begin{bmatrix}B\\ 0\end{bmatrix}u \qquad y=Cx$$

$$\begin{bmatrix}\dot{x}\\ \dot{e}\end{bmatrix}=\begin{bmatrix}0 & 0 & 0 & 0\\ 2 & -2 & 0 & 0\\ 0 & 0 & 0 & -8\\ 0 & 0 & 2 & -8\end{bmatrix}\begin{bmatrix}x\\ e\end{bmatrix}+\begin{bmatrix}1\\ -1\\ 0\\ 0\end{bmatrix}u \qquad y=[0\ 1\ 0\ 0]\begin{bmatrix}x\\ e\end{bmatrix}$$

$$\hat{A}\hat{B}=\begin{bmatrix}0 & 0 & 0 & 0\\ 2 & -2 & 0 & 0\\ 0 & 0 & 0 & -8\\ 0 & 0 & 2 & -8\end{bmatrix}\begin{bmatrix}1\\ -1\\ 0\\ 0\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\\ 0\end{bmatrix}$$

NOT controllable

$$CA=[0\ 1\ 0\ 0]\begin{bmatrix}0 & 0 & 0 & 0\\ 2 & -2 & 0 & 0\\ 0 & 0 & 0 & -8\\ 0 & 0 & 2 & -8\end{bmatrix}=[2\ -2\ 0\ 0]$$

NOT observable

c(d) z_1 does not alter the input-output response of the system, so $\frac{z_1}{u} = \frac{x_2}{u}$ and control inputs do not feedback into the state dynamics.

(c) Derive the combined state equation for the system plus observer. Take as state variables x_1, x_2, $e_1 = x_1 - z_1$, and $e_2 = x_2 - z_2$. Take u as input and z_1 as the output. Is this system controllable and/or observable? Give physical reasons for any states being uncontrollable or unobservable.

(d) What is the transfer function relating u to z_1? Explain your result.

$$C = \begin{bmatrix} 0 & 0 & 0 & 0.01 \\ 0 & 0 & 1 & -0.225 \\ 0 & 1 & -2.25 & -0.9044 \\ 1 & -2.25 & -90.4375 & 4.0336 \end{bmatrix} \quad \text{rank}(C) = 4 \quad \text{controllable}$$

$$A - Bk = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 500 & -15 & -95.5 & -2.25 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} k_1 & k_2 & k_3 & k_4 \end{bmatrix}$$

$$A - Bk = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 500 - k_1 & -15 - k_2 & -95.9 - k_3 & -2.25 - k_4 \end{bmatrix}$$

$$\lambda I - (A - Bk) = \begin{bmatrix} \lambda & -1 & 0 & 0 \\ 0 & \lambda & -1 & 0 \\ 0 & 0 & \lambda & -1 \\ k_1 - 500 & k_2 + 15 & k_3 + 95.9 & \lambda + k_4 + 2.25 \end{bmatrix}$$

$$\det = \lambda \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ k_2+15 & k_3+95.9 & \lambda + k_4 + 2.25 \end{vmatrix} + \begin{vmatrix} 0 & -1 & 0 \\ 0 & \lambda & -1 \\ k_1 - 500 & k_3 + 95.9 & \lambda + k_4 + 2.25 \end{vmatrix}$$

$$= \lambda \left(\lambda \begin{vmatrix} \lambda & -1 \\ k_3+95.9 & \lambda + k_4 + 2.25 \end{vmatrix} + \begin{vmatrix} 0 & -1 \\ k_2+15 & \lambda + k_4 + 2.25 \end{vmatrix} \right) + \begin{vmatrix} 0 & -1 \\ k_1 - 500 & \lambda + k_4 + 2.25 \end{vmatrix}$$

$$= \lambda^2 \left(\lambda^2 + k_4 \lambda + 2.25\lambda + k_3 + 95.9 \right) + \lambda (k_2 + 15) + k_1 - 500$$

$$= \lambda^4 + k_4 \lambda^3 + 2.25 \lambda^3 + (k_3 + 95.9) \lambda^2 + \lambda (k_2 + 15) + k_1 - 500$$

$$= \lambda^4 + (k_4 + 2.25) \lambda^3 + (k_3 + 95.9) \lambda^2 + (k_2 + 15) \lambda + k_1 - 500$$

$$\phi_{des} = s^4 + 222 s^3 + 12442 s^2 + 12440 s + 24000$$

$k_4 = 222 - 2.25 = 219.75$ $\quad k_3 = 12442 - 95.9$

$k_2 = 12440 - 15 = 12520.9$ $\quad k_1 = 24000 + 500 = 24500$

$k = [24500 \quad 12520.9 \quad 1244.2 \quad 219.75]$

$$C' = \begin{bmatrix} \cdot & \cdot & \cdot & - \\ \cdot & \cdot & - & \cdot \\ \cdot & - & \cdot & \cdot \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$b_1 = [0\ 0\ 0\ 1] C^{-1} = [1\ 0\ 0\ 0]$

$b_2 = b_1 A = [0\ 1\ 0\ 0] \quad b_3 = b_2 A = [0\ 0\ 1\ 0] \quad b_4 = b_3 A = [0\ 0\ 0\ 1]$

$P = T_4 \qquad k = k_c P$

$k = [24500 \quad 24425 \quad 12346.5 \quad 219.75]$

$$A' = A - Bk = A - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} k_4$$

$P_d = (s-p_1)(s-p_2)(s+100)(s+120) = (s^2 - (p_1+p_2)s + p_1p_2)(s+100)(s+120)$

$= (s^2 + 2s + 2)(s+100)(s+120) = (s^3 + 2s^2 + 2s + 100s^2 + 200s + 200)(s+120)$

$= (s^3 + 102s^2 + 202s + 200)(s+120) = s^4 + 102s^3 + 202s^2 + 200s + 120s^3 + 12240s^2 + 24240s + 24000$

$= s^4 + 222s^3 + 12442s^2 + 12440s + 24000$

$\beta_3 = 222 \quad \beta_2 = 12442 \quad \beta_1 = 12440 \quad \beta_0 = 24000$

$k_c = [\beta_0 - a_0 \quad \beta_1 - a_1 \quad \beta_2 - a_2 \quad \beta_3 - a_3] = [24000+500 \quad 12440-15 \quad 12442-95.5 \quad 222-2.25]$

$k_c = [24500 \quad 24425 \quad 12346.5 \quad 219.75]$

$a_0 = -500 \quad a_1 = 15 \quad a_2 = 95.5 \quad a_3 = 2.25$

(c) Select a desired set of poles that will yield
(i) a percent overshoot of less than 5%
(ii) a settling time of no more than 4.6 seconds
Hint: You will only get two poles to match your transient specs. You need to choose a spot for all closed-loop poles

$$C = [B \quad AB \quad A^2B \quad A^3B]$$

$$AB = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -2.25 \end{bmatrix} \quad A^2B = \begin{bmatrix} 0 \\ 1 \\ -2.25 \\ -90.4375 \end{bmatrix} \quad A^3B = \begin{bmatrix} 0.01 \\ -0.225 \\ -0.9044 \\ 4.0336 \end{bmatrix}$$

$$C = \begin{pmatrix} 0 & 0 & 0 & 0.01 \\ 0 & 0 & 1 & -0.225 \\ 0 & 1 & -2.25 & -0.9044 \\ 1 & -2.25 & -90.4375 & 4.0336 \end{pmatrix} \quad C^{-1} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$b_1 = [0\ 0\ 0\ 1]C^{-1} = [1\ 0\ 0\ 0]$$

$$b_2 = b_1A = [0\ 1\ 0\ 0] \quad b_3 = b_2A = [0\ 0\ 1\ 0] \quad b_4 = b_3A = [0\ 0\ 0\ 1]$$

$$P = I_4 \qquad K = K_C P$$

$$K = [24500 \quad 24425 \quad 12346.5 \quad 219.75]$$

Table of Contents

..... 1
Prob 1 1
Prob 2 2
prob 2 part 5 3
Prob 3 5


```
clc
clear all
close all

s = tf("s");
RAD2DEG = 180/pi;
DEG2RAD = pi/180;

prob = 3;


switch prob
```

Prob 1

```
case 1

% part 1
G = 1/ (s^2 + 2*s + 100) / (s^2 + 0.25*s - 5);

A = [0, 1, 0, 0;
     0, 0, 1, 0;
     0, 0, 0, 1;
     500, -15, -95.5, -2.25];
B = [0 0 0 1]';
C = [1 0 0 0];
D = 0;
G2 = ss(A,B,C,D);

% bode(G, G2);

% part 3.i
ts = 4.6;
mp = 0.05;
theta = atand(-pi/log(mp));
eta = cosd(theta);
wn = 4.6/eta/ts;
sigma = eta*wn;
wd = wn*sqrt(1 - eta*eta);

p1 = -sigma + 1i;
p2 = -sigma - 1i;
```

```
p3 = -100;
p4 = -120;

pd = (s-p1)*(s-p2)*(s-p3)*(s-p4);

pd_poly = pd.Numerator{1};
p_poly = G.Denominator{1};

d = pd_poly - p_poly;
k4 = d(2);
k3 = d(3);
k2 = d(4);
k1 = d(5);
K  = [k1 k2 k3 k4];

A_bar = A-B*K;
G_new = tf(ss(A_bar,B,C,D));
stepinfo(G_new)

lambdas = eig(A_bar)
fprintf('Pole %d: %9.4f %+.4fj\n', [(1:numel(lambdas)); real(lambdas).';
imag(lambdas).'])

step(G_new)
```

Prob 2

```
case 2

% w0 = 1;
% w0 = 1.2;
w0 = 1;

A = [0 1; w0^2 0];
B = [0 -1]';
C = [2 1];
D = 0;

x0 = [.5 1]';

% % 2.a
% Q = [1 0; 0 1];
% R = 1;

% % 2.b
Q = [9 0; 0 1];
R = 1;

% 2.c
% Q = [160 0; 0 1];
% R = 50;
```

```
[P,K,L] = icare(A,B,Q,R);      % L = closed-loop poles, G = K (state
feedback)                   % u = -K x
Acl = A - B*K;
K
eig(Acl);                       % should match L


% Plot
sys_cl = ss(Acl,B,C,D);

s = stepinfo(sys_cl)

% Sim
t = 0:0.01:5;
% r = ones(size(t));  % Unit step
r = zeros(size(t));  % Not unit step
[y, t, x] = lsim(sys_cl, r, t, x0);

u = -K*x';
u(1)

figure;
hold on;
plot(t, x(:,1), 'LineWidth', 1.5);
plot(t, x(:,2), 'LineWidth', 1.5); hold off;
xlabel('Time (s)');
legend('X1', 'X2');
grid on;
title('Output with no reference');


figure;
plot(t, u, 'r--', 'LineWidth', 1.5);
xlabel('Time (s)');
legend('u');
grid on;
title('Control input');
```

prob 2 part 5

```
case 2.2

B = [0 -1]';
C = [2 1];
D = 0;

x0 = [.5 1]';

% % 2.b
Q = [9 0; 0 1];
R = 1;

% 2.c
```

```
Q = [160 0; 0 1];
R = 50;

% First part
w0 = 0.8;
A = [0 1; w0^2 0];
[P,K,L] = icare(A,B,Q,R);      % L = closed-loop poles, G = K (state
feedback)                    % u = -K x
Acl = A - B*K;

% Plot
sys_cl_08 = ss(Acl,B,C,D);

% Sim
t = 0:0.01:5;
% r = ones(size(t));  % Unit step
r = zeros(size(t));  % Not unit step
[y_08, t_08, x_08] = lsim(sys_cl_08, r, t, x0);

u_08 = -K*x_08';

w0 = 1.2;
A = [0 1; w0^2 0];
[P,K,L] = icare(A,B,Q,R);      % L = closed-loop poles, G = K (state
feedback)                    % u = -K x
Acl = A - B*K;

% Plot
sys_cl_12 = ss(Acl,B,C,D);

% Sim
t = 0:0.01:5;
% r = ones(size(t));  % Unit step
r = zeros(size(t));  % Not unit step
[y_12, t_12, x_12] = lsim(sys_cl_12, r, t, x0);

u_12 = -K*x_12';

figure;
hold on;
plot(t, x_08(:,1), 'LineWidth', 1.5);
plot(t, x_08(:,2), 'LineWidth', 1.5);
plot(t, x_12(:,1), 'LineWidth', 1.5);
plot(t, x_12(:,2), 'LineWidth', 1.5); hold off;
xlabel('Time (s)');
legend('x1 (w0 = 0.8 rad/s)', 'x2 (w0 = 0.8 rad/s)', 'x1 (w0 = 1.2 rad/s)',
'x2 (w0 = 1.2 rad/s)');
grid on;
title('Problem b state response');

s_08 = stepinfo(sys_cl_08)
s_12 = stepinfo(sys_cl_12)
```

Prob 3

```
case 3


A = [-4 1; -2 -1];
B = [0 1; 1 2]';
C = [1 0; 3 1];
D = [0 0; 0 0];

x0 = [1 1]';
rho = 2;


Qy = [2 0; 0 1];
R = rho * [1 0; 0 2];

Qx = C'*Qy*C;

K_lqr = lqr(A, B, Qx, R);  % Compute the LQR gain
[P,K,L] = icare(A,B,Qx,R);
A_lqr = A - B * K;    % Closed-loop system with LQR
sys_cl = ss(A_lqr, B, C, D);  % State-space representation

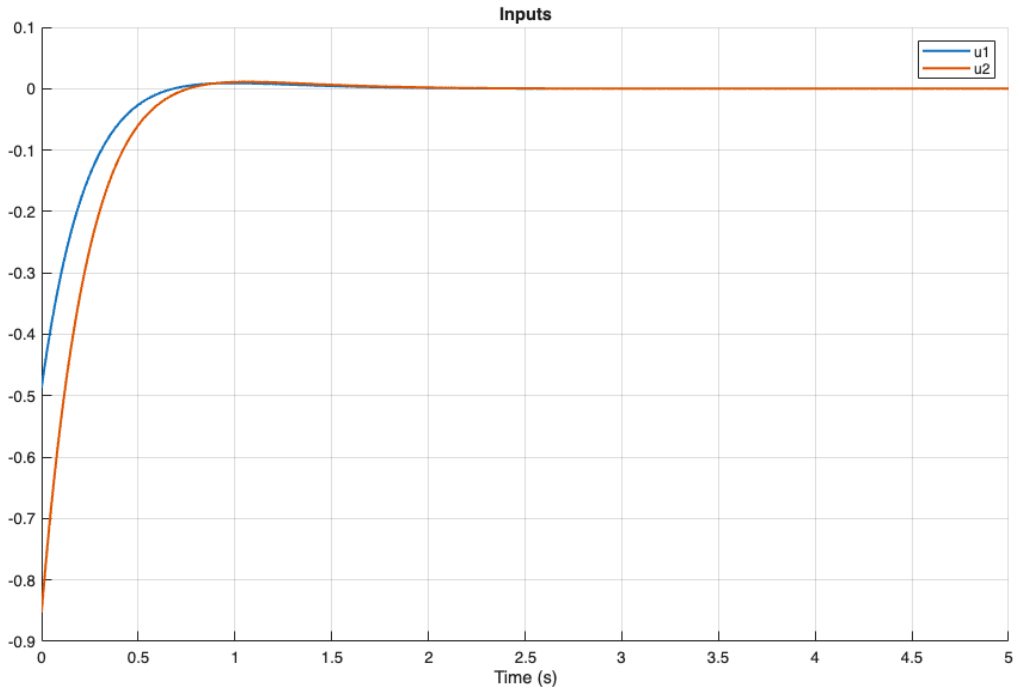

% Sim
t = 0:0.01:5;
N = 5/.01 + 1;
% r = ones(size(t));  % Unit step
r = zeros(2, N);  % Not unit step
[y, t, x] = lsim(sys_cl, r, t, x0);

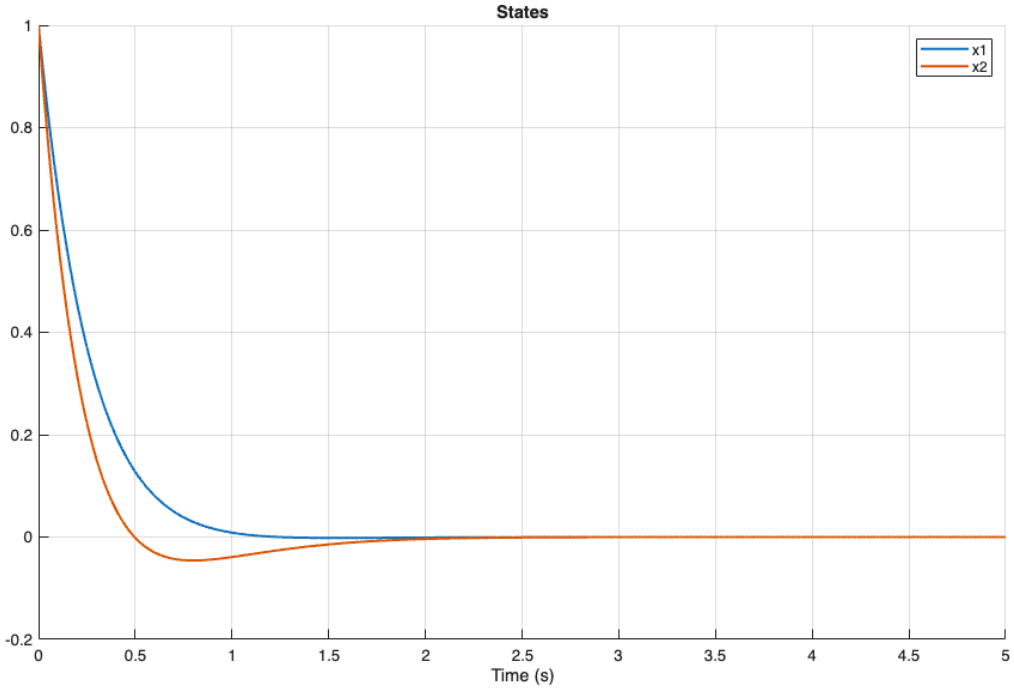
u = -K*x';

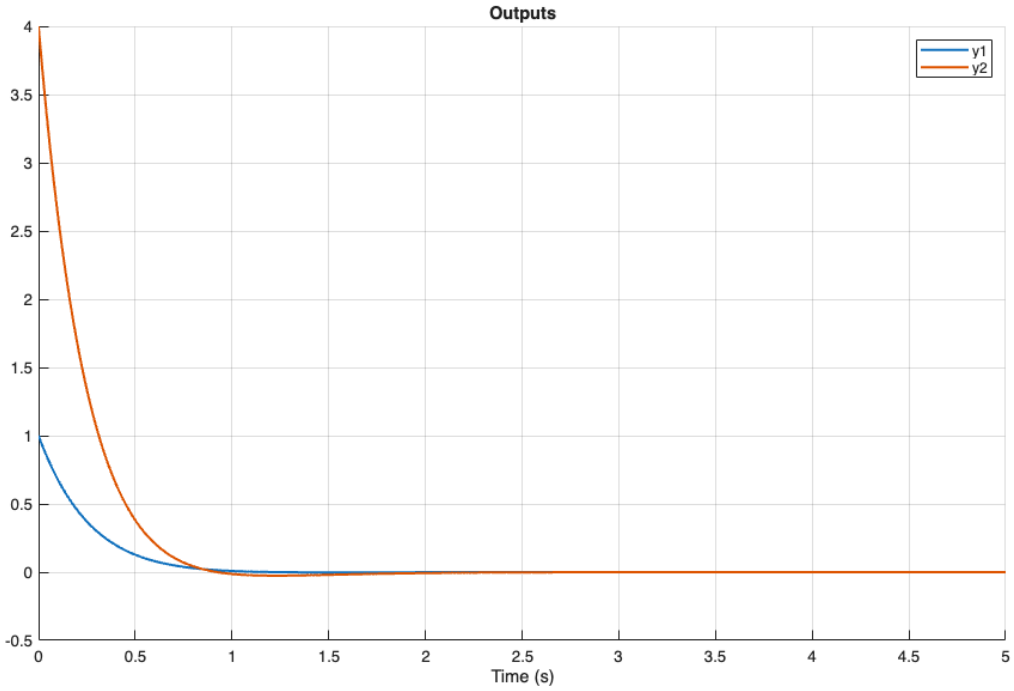

figure;
hold on;
plot(t, u(1,:), 'LineWidth', 1.5);
plot(t, u(2,:), 'LineWidth', 1.5);
xlabel('Time (s)');
legend('u1', 'u2');
grid on;
title('Inputs');

figure;
hold on;
plot(t, x(:,1), 'LineWidth', 1.5);
plot(t, x(:,2), 'LineWidth', 1.5);
xlabel('Time (s)');
legend('x1', 'x2');
grid on;
title('States');
```

```
figure;
hold on;
plot(t, y(:,1), 'LineWidth', 1.5);
plot(t, y(:,2), 'LineWidth', 1.5);
xlabel('Time (s)');
legend('y1', 'y2');
grid on;
title('Outputs');
end
```







Published with MATLAB® R2025b